HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 07 Name: _____ Marks: / 118

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. It is given that $g(x)$ is partly constant and partly varies as x. Suppose that $g(-3) = -21$ and $g(7) = 9$.

   (a) Find $g(x)$. (3 marks)

   (b) Let $h(x) = xg(x) + k$, where k is a real constant. If all the roots of the equation $h(x) = 0$ are real numbers, find the range of values of k. (3 marks)

   (6 marks) (DSE-2024 PI Q10)

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2. The stem-and-leaf diagram below shows the distribution of the numbers of hours spent on reading journals in a month by a group of researchers.

Stem (tens)	Leaf (units)
2	0 0 1 a a a 8 8 9 9
3	0 0 2 3 4 4 7 9
4	0 b

The mean of the distribution is 30 .

(a) Find a and b . (3 marks)

(b) Write down the least possible range of the distribution. (1 mark)

(c) Find the greatest possible inter-quartile range of the distribution. (3 marks)

(7 marks) (DSE-2024 PI Q11)

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3. Denote the origin by O.

(a) A and B are points lying on the positive x-axis such that the x-coordinate of A is greater than the x-coordinate of B. A vertical line which passes through B cuts the straight line $y = mx$ at the point C such that $AB = BC$, where m is a positive constant. Let D be a point such that $ABCD$ is a square. Express the slope of OD in terms of m. (3 marks)

(b) The coordinates of the points M and N are $(6, 5)$ and $(10, 0)$ respectively. Let P and Q be points lying on OM and MN respectively while R and S be points lying on the x-axis. If the quadrilateral $PQRS$ is a square, find the x-coordinate of P. (4 marks)

(7 marks) (DSE-2024 PI Q12)

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4. The base of a solid right pyramid is a square of side 64 cm. The height of the pyramid is 24 cm. The pyramid is divided into a frustum X and a pyramid Y by a plane which is parallel to its base. It is given that the height of Y is 18 cm.

(a) Find the volume of X. (3 marks)

(b) The base of another solid right pyramid is a square. This pyramid is divided into a frustum Z and a pyramid by a plane which is parallel to its base. The height and the total surface area of Z are 3 cm and 960 cm^2 respectively. Are X and Z similar? Explain your answer. (4 marks)

(7 marks) (DSE-2024 PI Q13)

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5. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

(a) Write down the value of p. (1 mark)

(b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

(i) Find q and r.

(ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

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6. It is given that $\log_9 y$ is a linear function of $\log_3 x$. Denote the graph of the linear function by L. The slope of L is 4 and L passes through the point $(5, 22)$. Express y in terms of x. (3 marks) (DSE-2024 PI Q15)

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7. In a bag, there are 16 red cups and 4 white cups. If 5 cups are randomly drawn from the bag at the same time, find

(a) the probability that exactly 1 white cup is drawn; (2 marks)

(b) the probability that at most 3 red cups are drawn. (2 marks)

(4 marks) (DSE-2024 PI Q16)

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8. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100 ? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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9. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

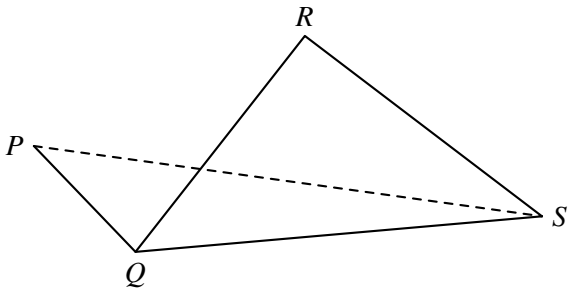


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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10. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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11. $\sqrt{333} =$

A. 18 (correct to the nearest integer).

B. 18.24 (correct to 2 decimal places).

C. 18.248 (correct to 3 significant figures).

D. 18.2482 (correct to 4 decimal places).

(2 marks) (DSE-2024 PII Q04)

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12. Let m be a constant. Solve the equation $x^2 - 3x = (m-1)^2 - 3(m-1)$.

A. $x = m - 1$ or $x = m - 4$

B. $x = m - 1$ or $x = 4 - m$

C. $x = 1 - m$ or $x = m - 4$

D. $x = 1 - m$ or $x = 4 - m$

(2 marks) (DSE-2024 PII Q07)

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13. Let $f(x) = x^3 + kx^2 + 5x + 10$, where k is a constant. If $f(x)$ is divisible by $x + k$, find the remainder when $f(x)$ is divided by $x + 1$.

A. -2

B. 2

C. 6

D. 18

(2 marks) (DSE-2024 PII Q09)

Answers written in the margins will not be marked.

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14. Which of the following statements about the graph of $y = 2(6-x)^2 - 7$ is true?

A. The graph opens upwards.

B. The graph does not cut the x-axis.

C. The y-intercept of the graph is -7.

D. The graph passes through the point $(-6, -7)$.

(2 marks) (DSE-2024 PII Q14)

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15. The ratio of the height of a right circular cylinder to the height of a right circular cone is 32 : 15 while the ratio of the volume of the circular cylinder to the volume of the circular cone is 10 : 9. If the base radius of the circular cylinder is 25 cm, then the base radius of the circular cone is

A. 20 cm.

B. 24 cm.

C. 48 cm.

D. 60 cm.

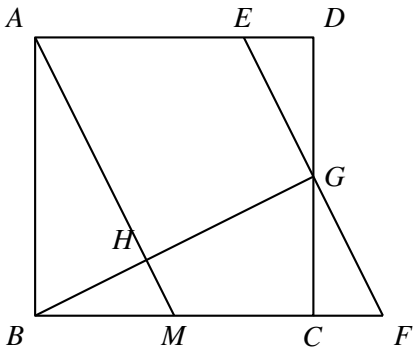
(2 marks) (DSE-2024 PII Q16)

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16. In the figure, $ABCD$ is a square. Let M be the mid-point of BC. E is a point lying on AD such that $AE : ED = 3 : 1$. F is a point lying on BC produced such that $EF//AM$. CD and EF intersect at the point G while AM and BG intersect at the point H. If the area of $\triangle BMH$ is $4\,\text{cm}^2$, then the area of the trapezium $AEGH$ is



A. $12\,\text{cm}^2$.

B. $33\,\text{cm}^2$.

C. $39\,\text{cm}^2$.

D. $45\,\text{cm}^2$.

(2 marks) (DSE-2024 PII Q17)

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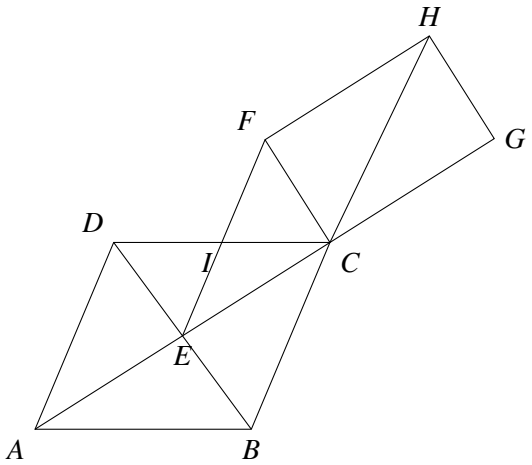
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17. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

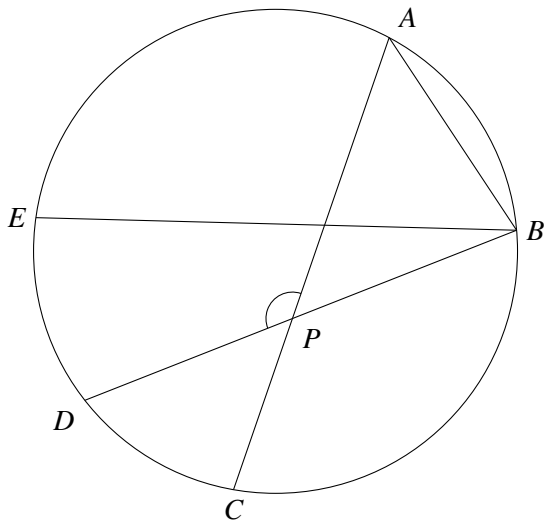
(2 marks) (DSE-2024 PII Q21)

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18. In the figure, $ABCD$ is a circle. AC and BE are diameters of the circle. Let P be the point of intersection of AC and BD. If $\angle ABE = 46°$ and $\angle DBE = 16°$, then $\angle APD =$



A. 104°.

B. 108°.

C. 120°.

D. 135°.

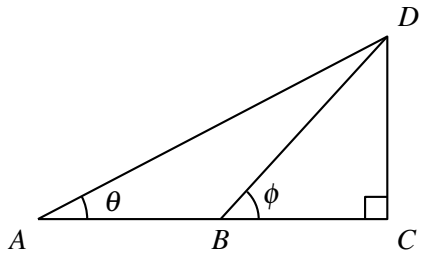
(2 marks) (DSE-2024 PII Q22)

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19. In the figure, ABC is a straight line. $\dfrac{BC}{AD} =$



A. $\dfrac{\sin\theta}{\tan\phi}$

B. $\dfrac{\tan\phi}{\sin\theta}$

C. $\dfrac{\cos\theta}{\tan\phi}$

D. $\dfrac{\tan\phi}{\cos\theta}$

(2 marks) (DSE-2024 PII Q23)

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20. The coordinates of the point U are $(-3, -8)$. U is rotated anticlockwise about the origin through $90°$ to the point V. V is then reflected with respect to the straight line $x = 2$ to the point W. Find the x-coordinate of W.

A. -4

B. -3

C. 7

D. 12

(2 marks) (DSE-2024 PII Q24)

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21. The coordinates of the points A and B are $(-3, 1)$ and $(-7, -5)$ respectively. If P is a point lying on the straight line $x - y + 13 = 0$ such that $AP = PB$, then the y-coordinate of P is

A. -11.

B. -2.

C. 2.

D. 11.

(2 marks) (DSE-2024 PII Q25)

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22. Find the constant k such that the straight lines $6x - 8y = 7k$ and $kx + 12y = 5$ do not intersect with each other.

A. -16

B. -9

C. 9

D. 16

(2 marks) (DSE-2024 PII Q26)

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23. Two numbers are randomly drawn at the same time from six cards numbered 1, 2, 3, 4, 5 and 6 respectively. Find the probability that the product of the numbers drawn is not less than 12.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{7}{15}$

D. $\frac{8}{15}$

(2 marks) (DSE-2024 PII Q28)

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24. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188$.

B. $192 \times 16^{11} + 205$.

C. $175 \times 16^{12} + 188$.

D. $192 \times 16^{12} + 205$.

(2 marks) (DSE-2024 PII Q32)

Answers written in the margins will not be marked.

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Answers written in the margins will not be marked.

25. If $x = \log_2 y - 2$ and $(\log_2 y)^2 = 5\log_2 y + x - 7$, then $y =$

A. 1.

B. 8.

C. 1 or 3.

D. 3 or 8.

(2 marks) (DSE-2024 PII Q33)

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26. Let $z = (a-5)i + \dfrac{(a+2)i}{2+i}$. If a and z are real numbers, then $a - z =$

A. 2.

B. 3.

C. 4.

D. 5.

(2 marks) (DSE-2024 PII Q35)

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27. The sum of the first n terms of a sequence is $n(2n+3)$. Which of the following are true?

I. 14 is a term of the sequence.

II. The nth term of the sequence is $4n+1$.

III. The sequence is an arithmetic sequence.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q36)

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28. Consider the following system of inequalities:

$$\begin{cases} x - 2y \leq 1 \\ x + 4y \leq 13 \\ 2x - y \geq -1 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. Find the constant c such that the least value of $5x - 2y + c$ is 22, where (x, y) is a point lying in R.

A. 1

B. 23

C. 25

D. 29

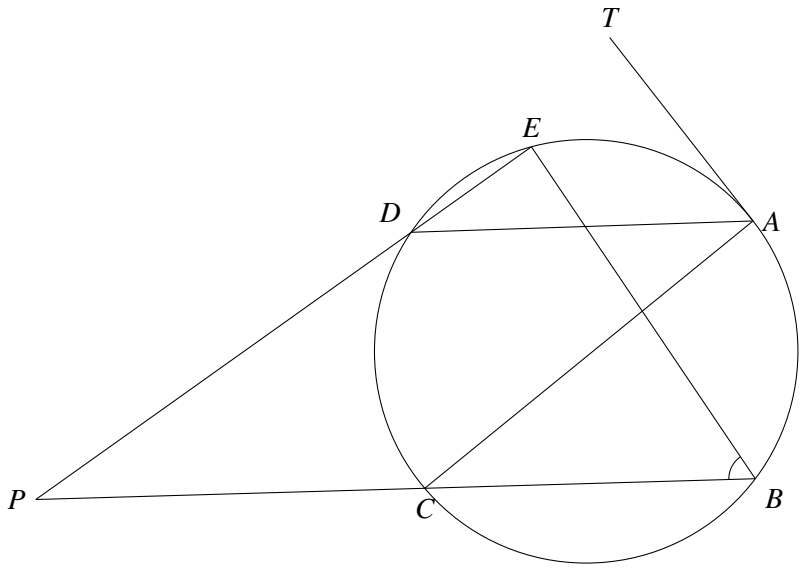
(2 marks) (DSE-2024 PII Q37)

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29. In the figure, TA is the tangent to the circle $ABCDE$ at the point A. BC produced and ED produced meet at the point P. If $\angle ACB = 43°$, $\angle DAT = 55°$ and $\angle CPD = 29°$, then $\angle CBE =$



A. 64°.

B. 69°.

C. 72°.

D. 78°.

(2 marks) (DSE-2024 PII Q38)

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30. For $0° < \theta \leq 360°$, how many roots does the equation $4\cos^2\theta - 3\cos\theta - 1 = 0$ have?

A. 3

B. 4

C. 5

D. 6

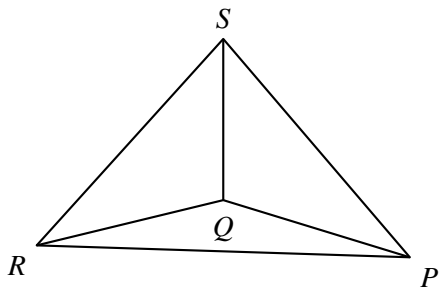
(2 marks) (DSE-2024 PII Q39)

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31. In the figure, the base PQR of the tetrahedron $PQRS$ lies on the horizontal ground. It is given that Q is vertically below S. If $\angle PQR = 90°$, $\angle QPS = 30°$ and $\angle QRS = 45°$, then $\cos\angle PRS =$



A. $\frac{1}{2}$.

B. $\frac{3}{4}$.

C. $\frac{\sqrt{2}}{4}$.

D. $\frac{\sqrt{3}}{6}$.

(2 marks) (DSE-2024 PII Q40)

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32. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

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33. A queue is formed by 2 managers and 7 officers. If no managers are next to each other, how many different queues can be formed?

A. 80 640

B. 141 120

C. 282 240

D. 362 880

(2 marks) (DSE-2024 PII Q42)

Answers written in the margins will not be marked.

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34. If the variance of the seven numbers x_1, x_2, x_3, x_4, x_5, x_6 and x_7 is 16, then the standard deviation of the seven numbers $9x_1 - 5$, $9x_2 - 5$, $9x_3 - 5$, $9x_4 - 5$, $9x_5 - 5$, $9x_6 - 5$ and $9x_7 - 5$ is

A. 31.

B. 36.

C. 139.

D. 144.

(2 marks) (DSE-2024 PII Q45)

I can finish _____ questions

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